



DPUH264Cap

Controls H.264 video capture devices that are manufactured by BlackmagicDesign (<http://www.blackmagicdesign.com>).

1. Parameters for naming the files:

<i>Path</i>	Output path of the recorded file. If no absolute path is provided, the path is located below SILAB's installation path. Default: "VIDEO\"
<i>FileName</i>	File name of the created video (within the Path folder). Default: "Video" Alternatively, it can be specified using the launch data <code>VAR_DATA-FILENAME</code> .
<i>Suffix</i>	Suffix that is inserted into the file name (after <code>FileName</code> , but before the extension ".avi").
<i>DiskFree_Warning_MB</i>	Minimum recommended free disk space (MB) before starting a simulation. If less space is free, a warning is issued. Default: 10000
<i>DiskFree_Error_MB</i>	Minimum required free disk space (MB) before starting a simulation. If less space is free, the simulation cannot be started, because the space will probably run out. Default: 500
<i>OverwriteExistingFile</i>	If set to true, an already existing video file is overwritten. If set to false (the default), an error message is issued and the simulation cannot be started if the output file already exists.
<i>LD_SaveDataToken</i>	The name of a LaunchData entry (of type <code>LD_BOOL</code>), which defines if a video file is created. Default: <code>VAR_SAVEDATA</code>
<i>LD_DataFileNameToken</i>	The name of a LaunchData entry (of type <code>LD_EDIT</code>), which defines the video file name (instead of the File-Name parameter). Default: <code>VAR_DATAFILENAME</code>

2. Parameters concerning video compression:

<i>EncoderPreset</i>	Preset used for the hardware H.264 encoder. Usually, the values 0..5 are valid; where 0 is the default preset (may be interlaced) and 1 is a non-interlaced preset. The possible presets are listed in the SILAB system log while the simulation is prepared. Default: 0
<i>PrepareDelay</i>	Number of seconds the DPU waits for the video capture device to initialize during preparation. Default: 5



SILAB DRIVING SIMULATION

version Version 7.1 documentation

3. Outputs:

VideoTimeMS

Current video time stamp, in milliseconds. If a DPUDData-File recording is running in parallel, it is recommended to add this variable to the DPUDDataFile, so that the files can be synchronized precisely afterwards.

NAudioPackets

Number of audio packets recorded so far.

NVideoPackets

Number of video packets recorded so far.

NTSPackets

Number of transport stream packets recorded so far.

FileNumber

Selected file number.

InputMode

Text describing which device input is recorded.

EncoderMode

Text describing the selected H.264 encoder setting.

EncoderInfo

Additional information about the H.264 encoder.

4. Functional principle:

Records digital video using a BlackmagicDesign H.264 capture device.

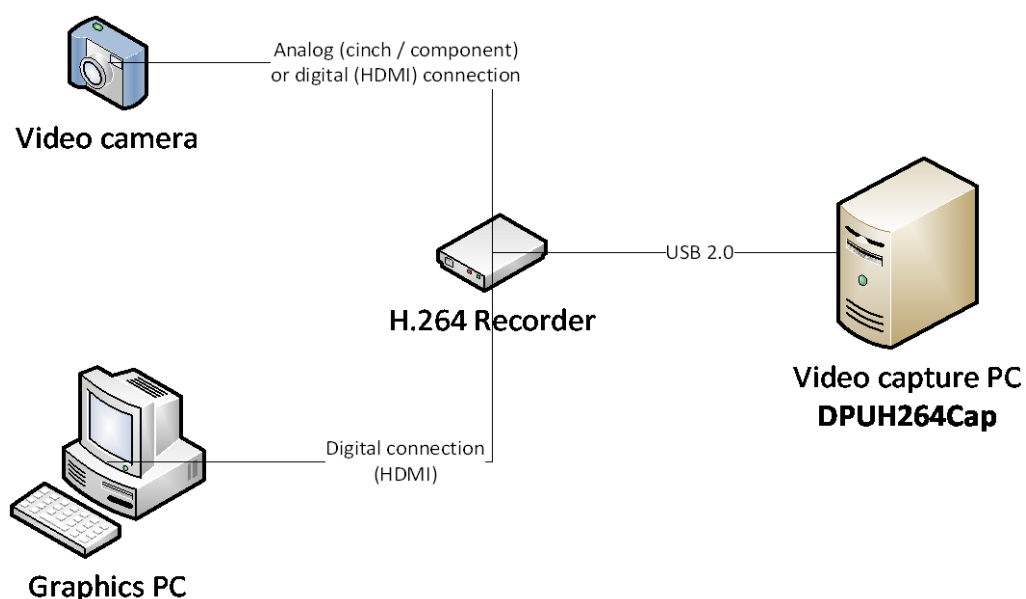
Compatible devices include the H.264 Pro Recorder (<http://www.blackmagicdesign.com/products/h264prorecorder>) and the ATEM Television Studio (<http://www.blackmagicdesign.com/products/atemtelevisionstudio>).

All of these devices support different analog and digital inputs (so that most types of video can be captured) and feature a builtin H.264 hardware video encoder, so that highquality digital video files can be captured without demands on the connected PC.

The captured video can come either from a video camera (using an analog or digital connection) or from another computer (using a HDMI connection). The H.264 recorder automatically detects which input is connected.

The output video is stored on a computer that is connected using USB 2.0.

The following graphic details the required connections.





Note: If the output of a graphics PC is recorded (for example, the simulator scenery rendered by DPUSGEView), please ensure that one of the supported resolutions is used. Usually, the TV formats 1280x720 pixels and 1920x1080 pixels can be used.

5. Example:

This DPU works well using the following script:

```
DPUH264Cap VideoCap
{
    Computer = {REC};
    Index = 90;
    Path = "\\Video\\MyProject\\";
};
```